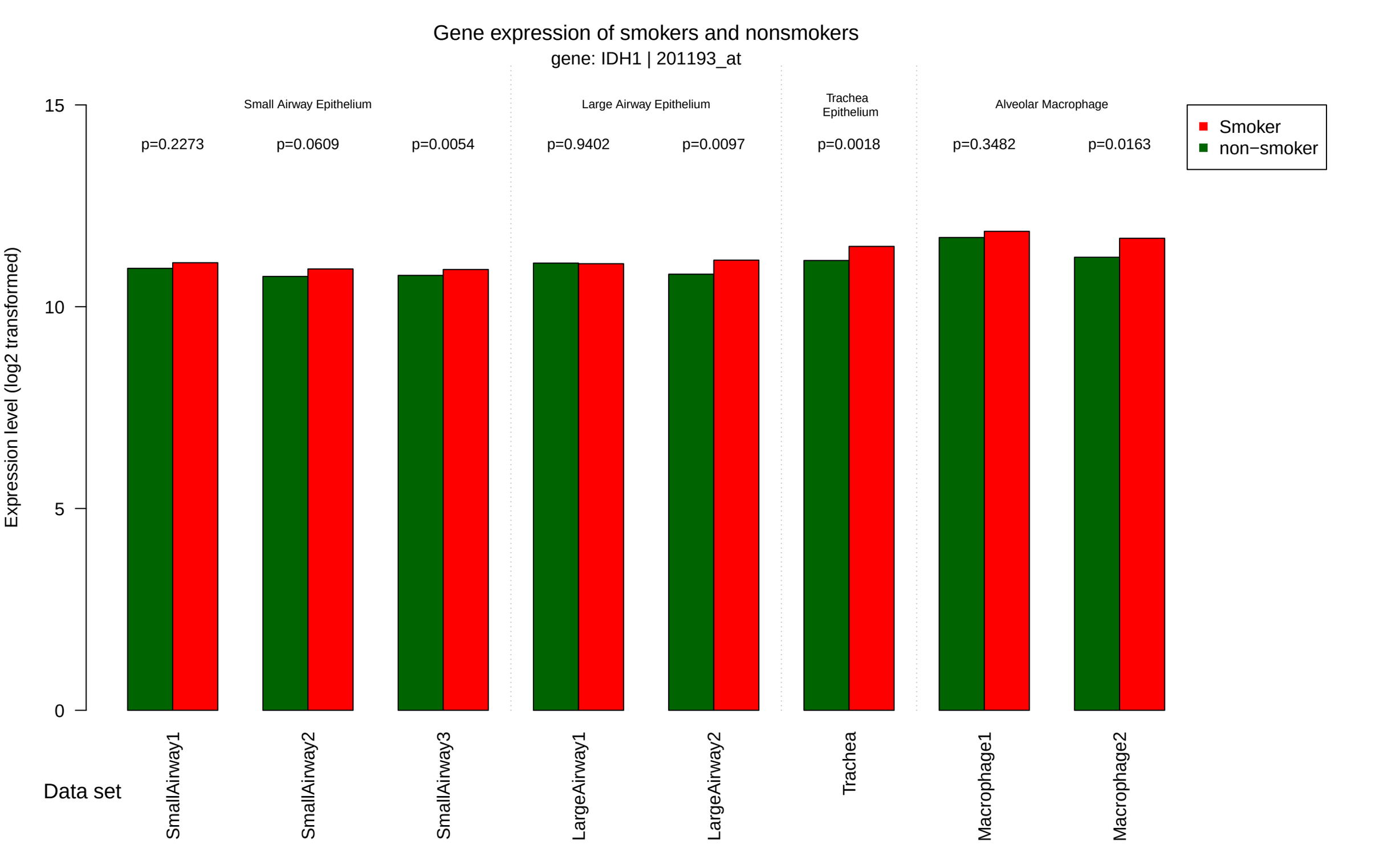
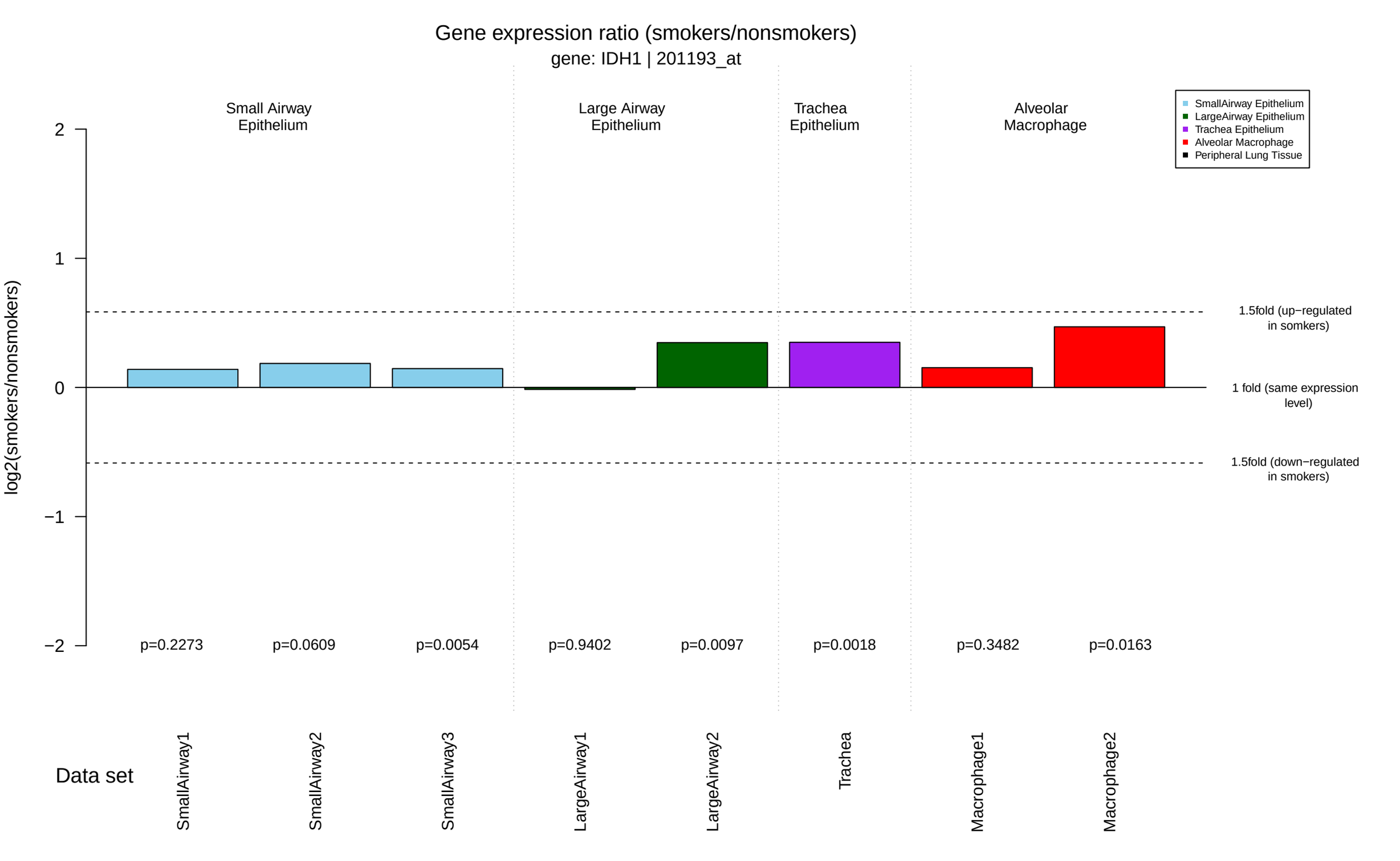
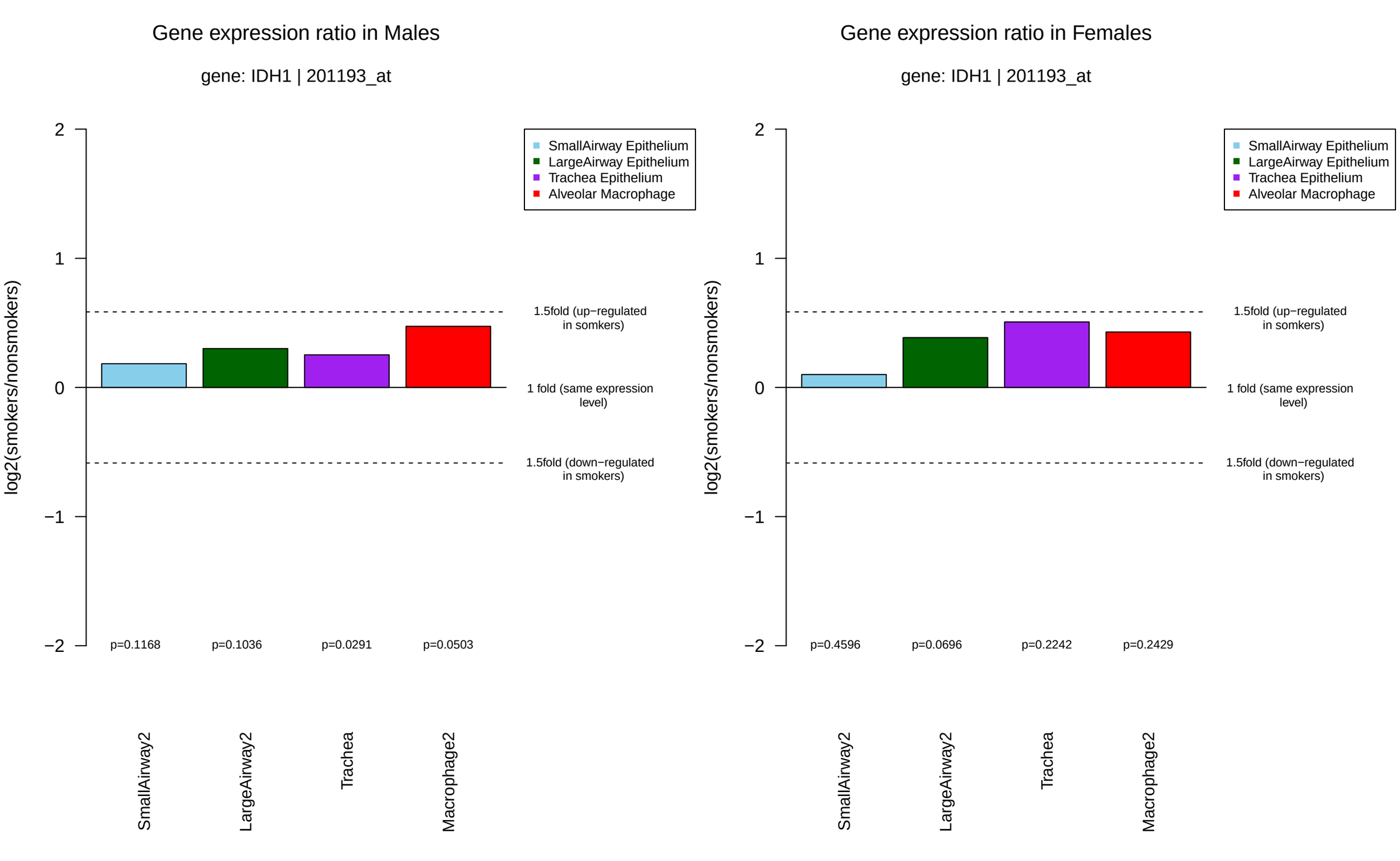
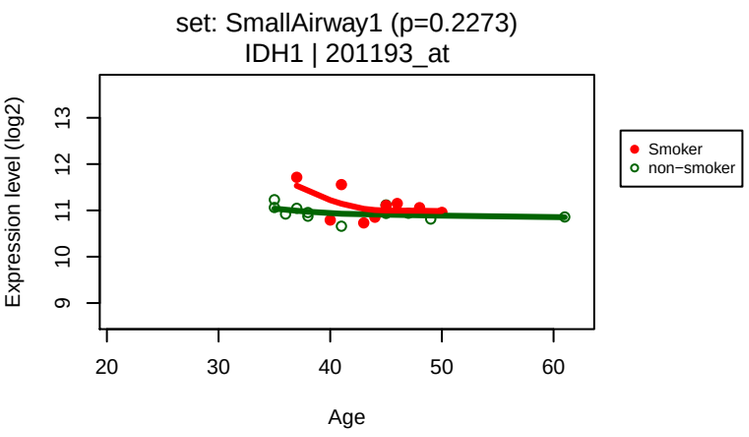
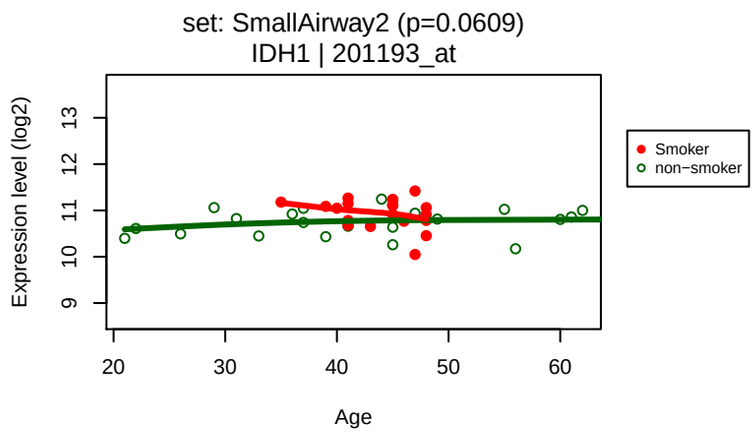
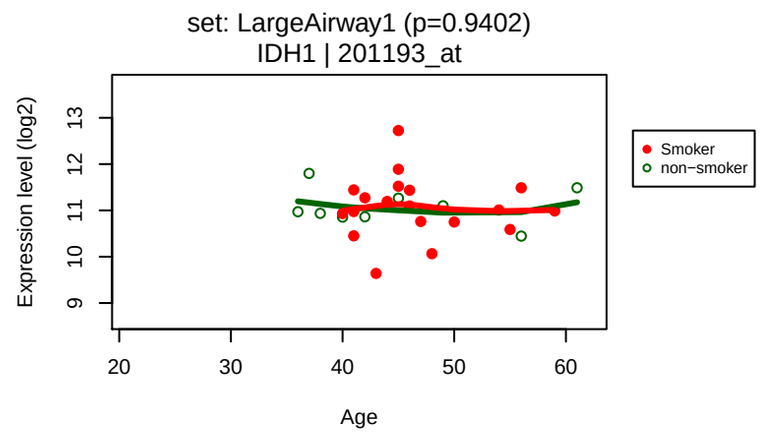

Gene expression of smokers and nonsmokers
gene: IDH1 | 201193_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2273
p=0.0609
p=0.0054
p=0.9402
p=0.0097
p=0.0018
p=0.3482
p=0.0163
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

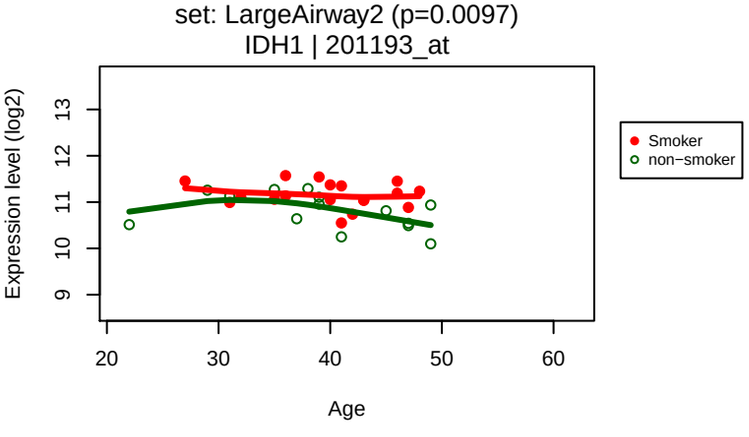

Gene expression ratio (smokers/nonsmokers)
gene: IDH1 | 201193_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2273
p=0.0609
p=0.0054
p=0.9402
p=0.0097
p=0.0018
p=0.3482
p=0.0163
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

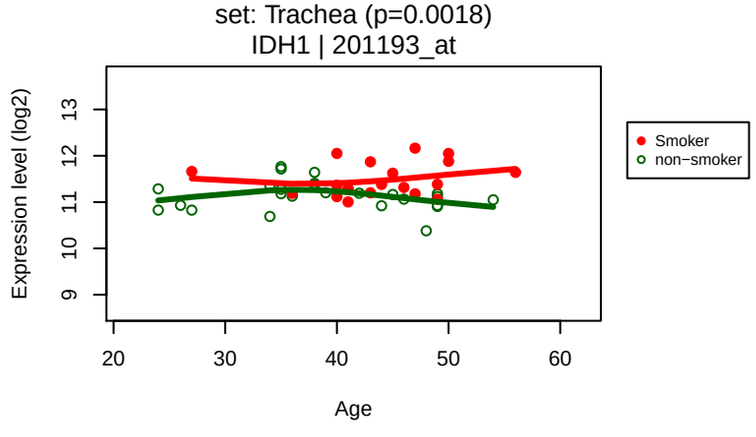

Gene expression ratio in Males
gene: IDH1 | 201193_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1168
p=0.1036
p=0.0291
p=0.0503
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IDH1 | 201193_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4596
p=0.0696
p=0.2242
p=0.2429
SmallAirway2
LargeAirway2
Trachea
Macrophage2

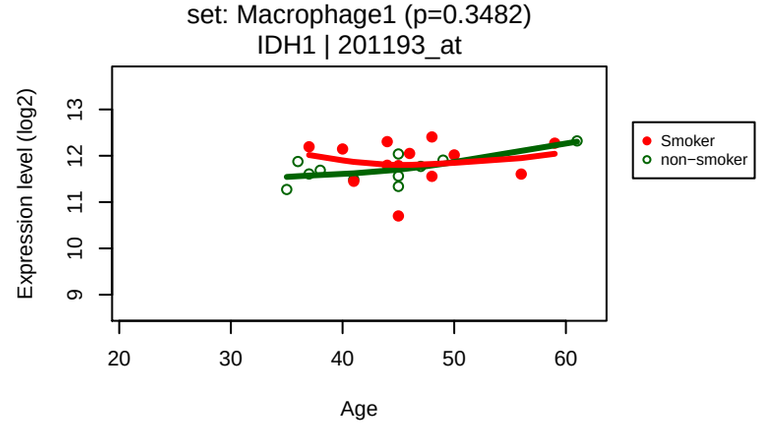

set: SmallAirway1 (p=0.2273)
IDH1 | 201193_at
Expression level (log2)
Age
Smoker
non−smoker

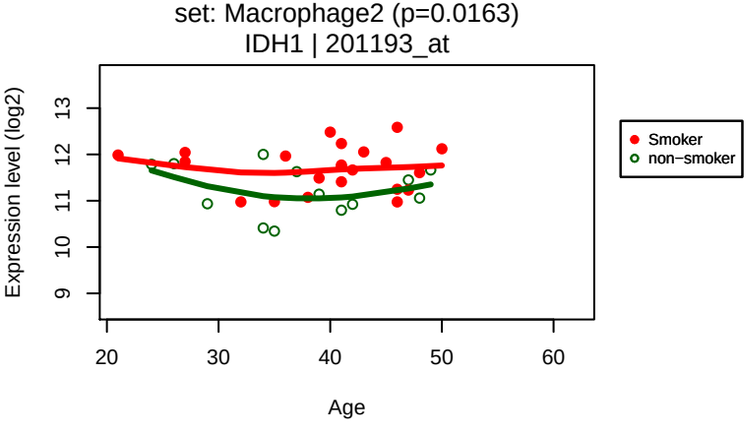

set: SmallAirway2 (p=0.0609)
IDH1 | 201193_at
Expression level (log2)
Age
Smoker
non−smoker

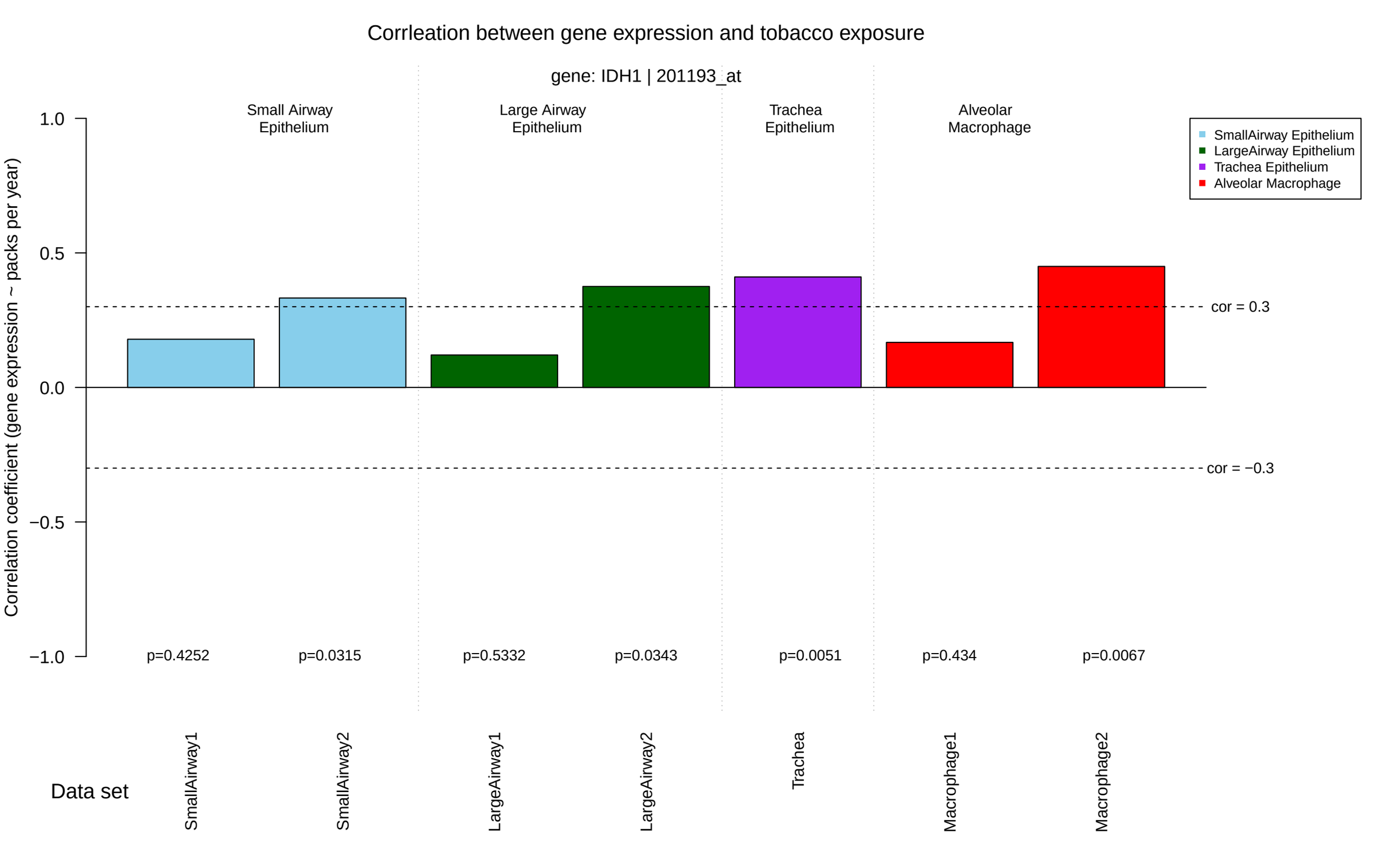

set: LargeAirway1 (p=0.9402)
IDH1 | 201193_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0097)
IDH1 | 201193_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0018)
IDH1 | 201193_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3482)
IDH1 | 201193_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0163)
IDH1 | 201193_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IDH1 | 201193_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4252
p=0.0315
p=0.5332
p=0.0343
p=0.0051
p=0.434
p=0.0067
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2